

## Lecture Goals

- Understand what a derivative really is
- Connects derivatives to slope & rates of change
- Learn how to compute derivatives using the limit definition
- Find derivatives of polynomial functions
- Practice w/ many examples & problems

## BIG PICTURE

The derivative of  $f$  @  $a$ , denoted by  $f'(a)$  measures how fast  $f(x)$  is changing @  $x=a$ . It is the slope of the tangent line to the graph of  $f$  @  $x=a$ .

## Motivating Remarks

### Recall Rates of Change

#### POSITION & VELOCITY

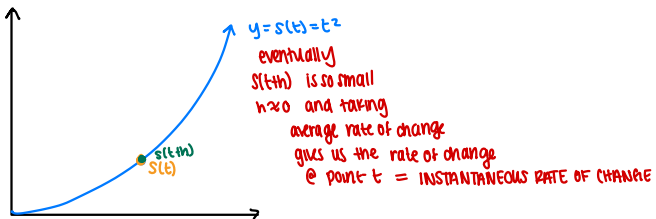
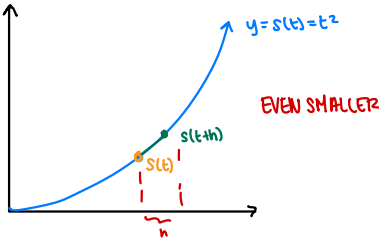
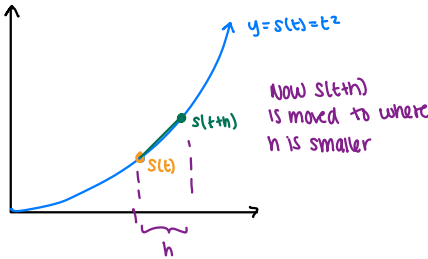
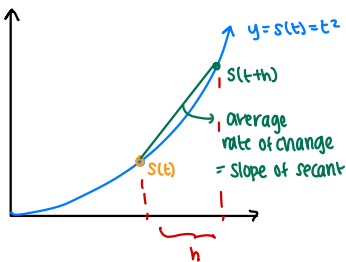
Suppose a car's position @ time  $t$  seconds is  $s(t) = t^2$  feet  
What's the average velocity on  $[t, t+h]$

### Recall formula for average rate of change [a,b]

$$\frac{f(b) - f(a)}{b - a} \quad \text{Apply Here } [t, t+h] \Rightarrow \frac{s(t+h) - s(t)}{t+h - t} = \frac{(t+h)^2 - t^2}{h} = \frac{t^2 + 2th + h^2 - t^2}{h} = \frac{2th + h^2}{h} = \frac{h(2t+h)}{h} = 2t+h$$

As  $h \rightarrow 0$  this approaches  $2t$

so our instantaneous velocity @ time  $t$ :  $v(t) = 2t$  ft/second



The derivative generalizes this idea to any function

# DERIVATIVE AS A LIMIT

Let  $f$  be a function. The derivative of  $f$  at  $a$  is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

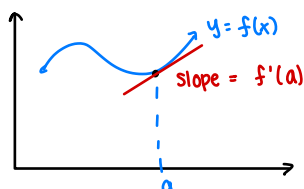
provided the limit exists.

- $h$  is a small change in  $x$
- $f(a+h) - f(a)$  is the change in  $f(x)$
- The ratio = average rate of change on  $[a, a+h]$
- As  $h \rightarrow 0$ , this becomes the instantaneous rate of change

We denote this as  $f'(a)$  = derivative of  $f$  at  $a$   
 $f'(x)$  = derivative function

## DERIVATIVE AS A SLOPE

Geometrically  $f'(a)$  = slope of the tangent line to the graph of  $f$  at  $x=a$

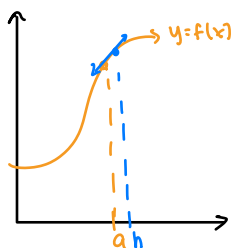
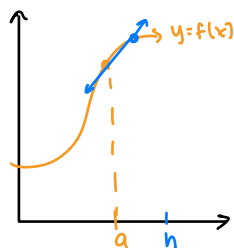


Secant line slope (average rate of change)

$$m_{\text{sec}} = \frac{f(a+h) - f(a)}{h}$$

Tangent line slope = instantaneous rate

$$m_{\text{tan}} = \lim_{h \rightarrow 0} m_{\text{sec}} = f'(a)$$



As  $h$  gets closer & closer to 0 the point  $(a+h, f(a+h))$  gets closer to  $(a, f(a))$

Secant line approaches the tangent

## DIFFERENTIABLE FUNCTIONS

def: A differentiable function is one whose derivative exists at every point in its domain.

Conditions

- 1)  $f(a)$  = defined
- 2)  $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$  exists and is finite (not  $\infty$  or  $-\infty$ )

# Examples

EXAMPLE 1:  $f(x) = x^2$  Find  $f'(a)$ .

use limit definition:

$$f'(a) = \lim_{h \rightarrow 0} \frac{\overset{f(a+h)}{(a+h)^2} - \overset{f(a)}{a^2}}{h} = \frac{a^2 + 2ah + h^2 - a^2}{h} = \frac{2ah + h^2}{h} = \frac{h(2a+h)}{h} = 2a+h$$

$$= 2a$$

$$f'(a) = 2a$$

EXAMPLE 2:  $f(x) = 3x+5$  Find  $f'(a)$   $a \neq 0$

$$f'(a) = \lim_{h \rightarrow 0} \frac{3(a+h)+5 - (3a+5)}{h} = \frac{3a+3h+5-3a-5}{h} = \frac{3h}{h} = 3$$

EXAMPLE 3:  $f(x) = \frac{1}{x}$  Find  $f'(a)$

$$f'(a) = \lim_{h \rightarrow 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h} = \lim_{h \rightarrow 0} \frac{\frac{a-(a+h)}{a(a+h)h}}{h} = \frac{a-a-h}{(a^2+ah)h} \cdot \frac{-h}{(a^2+ah)h} = \frac{-1}{a^2+ah} = -\frac{1}{a^2}$$

EXAMPLE 4:  $f(x) = \sqrt{x} = x^{1/2}$  Find  $f'(a)$  ( $a > 0$ )

$$f'(a) = \lim_{h \rightarrow 0} \frac{\sqrt{a+h} - \sqrt{a}}{h} \cdot \frac{\sqrt{a+h} + \sqrt{a}}{\sqrt{a+h} + \sqrt{a}} = \lim_{h \rightarrow 0} \frac{a+h-a}{h(\sqrt{a+h} + \sqrt{a})} = \frac{1}{\sqrt{a+h} + \sqrt{a}} = \frac{1}{2\sqrt{a}}$$

$$f'(a) = \frac{1}{2\sqrt{a}}$$

You try: a)  $f(x) = x^2 - 5x + 3$  b)  $f(x) = \frac{1}{x^2}$

## DERIVATIVE PROPERTIES

1) Derivative of a constant = 0

$$f(x) = 5$$

$$f'(x) = 0$$

2) POWER RULE

$$f(x) = x^n$$

$$f'(x) = nx^{n-1}$$

for any real  $n$

Recall Exponent Rules

$$a^{m/n} = \sqrt[n]{a^m} \quad \text{Ex } \sqrt{x} = x^{1/2}$$

$$a^{-m} = \frac{1}{a^m} \quad \text{Ex } \frac{1}{x^2} = x^{-2}$$

Example:

$$f(x) = x^3$$

$$f'(x) = 3x^{3-1} = 3x^2$$

Example:

$$f(x) = \frac{1}{12} x^4 = \frac{4}{12} x^3 = \frac{1}{3} x^3$$

You try:  $f(x) = \sqrt{x}$  &  $\frac{1}{x^2}$

## Derivative of Polynomial

A polynomial has the form:

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

where  $a_n, \dots, a_0$  are constants

Notice: Polynomials

are  $x^n$  added to each other

↳ taking derivative means just applying power to each term.

Derivative of a Polynomial

$$P'(x) = n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \dots + a_1$$

constant term  $a_0$  disappears

**Example:** Find the derivative

$$P(x) = 4x^3 - 3x^2 + 5x - 7$$

$$P'(x) = 12x^2 - 6x + 5$$

Why? Differentiate term by term using power rule

$$(4x^3)' = 12x^2 \quad (3x^2)' = 6x$$

$$(5x)' = 5 \quad (-7)' = 0$$

**Example:**

$$f(x) = x^2 - 5x + 3$$

$$f'(x) = 2x - 5$$

YOU TRY:

$$1) f(x) = 6x$$

$$2) f(x) = 4x^3 - x^2 + 2x - 1$$

$$3) f(x) = 5x^4 + 2x^2 - 3$$

$$4) f(x) = \sqrt[3]{x^5}$$

$$5) f(x) = (3x^2 + 1)(x - 4)$$